

Nitant Suhagiya

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Education

Toronto Metropolitan University

BSc. Computer Science with Minor in Mathematics

Dean's List 2025

Toronto, ON

Sept 2024 – May 2027

University of Regina

Diploma in Computer Science, **GPA 7.4/9**

Regina, SK

May 2019 – June 2021

Experience

Data Engineer Intern

Canadian Nuclear Safety Commission (CNSC)

May 2026 – Present

Ottawa, ON

- Engineered a Python automation pipeline that translates and validates data inputs for reactor simulation software, reducing manual data preparation time by 70% and eliminating transcription errors across critical safety workflows.
- Processed and mapped 10,000+ structured records per run, enabling consistent, audit-ready data handoffs in a classified, federally regulated nuclear safety environment.

Senior Research Assistant

CMIC Lab, TMU

Feb 2026 – Present

Toronto, ON

Research Assistant

Oct 2025 – Feb 2026

- Collaborating with a multidisciplinary research team to study the social dynamics of computer-mediated communication (CMC).
- Analyzing behavioural and communication data using statistical tools to identify patterns related to deception detection and social prediction.

Projects

Pagenova | Chrome AI assistant Extension (React, TypeScript, Vite)

- Built a Chrome extension that grounds streaming AI chat in open tabs, uploaded documents, and images, supporting local Ollama and cloud providers (OpenAI, Anthropic, Gemini, OpenAI-compatible).
- Designed a budget-aware RAG pipeline (chunk → embed → retrieve) with on-device Transformers.js embeddings, IndexedDB vector storage, and automatic full-context vs. retrieval mode based on model window size.

F1 Winner Predictor | Python, scikit-learn, FastF1 API

- Preprocessed and analyzed free practice session data from the FastF1 API (driver speed, lap times, conditions) to build a structured dataset for predictive modelling.
- Trained an XGBoost model to forecast driver speeds and race performance for current-season events using last year's race data as benchmarks, improving prediction accuracy through feature engineering.

Research

UM-GCN: Multiscale Label-Attention Graph Network for Chest X-Ray Classification | (Submitted, SPIE Medical Imaging 2027)

- Designed UM-GCN, a multiscale label-attention graph network that builds image-specific disease graphs from attention overlap, replacing fixed co-occurrence statistics with entropy-modulated message passing.
- Achieved macro-AUROC of 0.8318 and macro-AP of 0.3226 on ChestX-ray14, outperforming a tuned EfficientNet-B4 baseline through controlled ablations, with statistically significant AUROC gains on 5 disease classes (Atelectasis, Infiltration, Pneumonia, Edema, Emphysema) confirmed via external validation on CheXpert.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C/C++, SQL, PHP, HTML/CSS

Frameworks: React, Node.js, Express, Tailwind, FastAPI, Scikit-Learn, XGBoost, TensorFlow, Keras, PyTorch, Pandas, NumPy, Matplotlib, ONNX, Transformer, MCP, RAG, JAX, NLP, Computer Vision, LLM

Developer Tools: Git, GitHub, VS Code, Linux/UNIX, AWS, Power BI, Hugging Face, Flask, Docker, Jupyter Notebook, LangChain, LangGraph